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# Principal Component Analysis

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# PLAN

- Introduction
- Type of data
- Method
- General Algorithm
- Conclusion



# 1. Introduction

## 1.1 Definition

Principal component analysis, abbreviated by its initials P.C.A, it is multidimensional data factorial methods. It describes the statistical units and observed variables based on the study of linear correlation coefficients.



## 1.2 Goals

- ✓ Form homogeneous groups of statistical units.
- ✓ Analyses the relationships between variables.

**We try to concretize the fundamental properties of the data, using numerical parameters and graphs**

## 2. Type of data

### 2.1 Initial data

The data for the PCA is usually presented in the form of a rectangular table of individuals in rows and quantitative variables in columns.

|            |           |             |       |       |     |
|------------|-----------|-------------|-------|-------|-----|
|            | VARIABLES |             |       |       |     |
|            | 1         | .....       | $k$   | ..... | $K$ |
| INDIVIDUUS | 1         | <div></div> |       |       |     |
|            | $\vdots$  |             |       |       |     |
|            | $\vdots$  |             |       |       |     |
|            | $\vdots$  |             |       |       |     |
|            | $i$       |             |       |       |     |
|            | $\vdots$  |             |       |       |     |
|            | .....     | $x_{ik}$    | ..... |       |     |
|            | $\vdots$  |             |       |       |     |
|            | $\vdots$  |             |       |       |     |
|            | $\vdots$  |             |       |       |     |
|            | $I$       |             |       |       |     |

the general term in the  $i^{\text{th}}$  row and the  $k^{\text{th}}$  column is noted  $x_{ik}$  and it is the observation of the variable  $k$  on the individual  $i$

*Figure 2.1: Data table*

## 2.2 Examples

**Sensory analysis:** score of descriptor  $k$  for product  $i$ .

**Ecology:** concentration of pollutant  $k$  in river  $i$ .

**Economics:** value of indicator  $k$  for year  $i$ .

**Biology:** measure  $k$  for animal  $i$ .

**Marketing:** satisfaction index value  $k$  for brand  $i$ .

## 2.3 Exemple

|    | weight | size | Age | score |
|----|--------|------|-----|-------|
| 1  | 45     | 1.50 | 13  | 14    |
| 2  | 50     | 1.60 | 13  | 16    |
| 3  | 50     | 1.65 | 13  | 15    |
| 4  | 60     | 1,75 | 15  | 9     |
| 5  | 60     | 1,70 | 14  | 10    |
| 6  | 60     | 1,70 | 14  | 7     |
| 7  | 70     | 1,60 | 14  | 8     |
| 8  | 65     | 1.60 | 13  | 13    |
| 9  | 60     | 1.55 | 15  | 17    |
| 10 | 65     | 1.70 | 14  | 11    |

*table.2 : example Data*

## Remarques

- The data processed by PCA must be quantitative (homogeneous or heterogeneous, discrete or continuous).
- PCA will only give interesting results on large tables (number of units statistique  $> 15$  and number of variables  $> 4$ ).
- we can introduce in the table of the data the additional (passive) to facilitate the interpretation of the results..



## 3. Method

### 3.1 Concept of distance

two statistical units are similar if the observed variables take similar values on these two statistical units.

|   |    |      |    |    |
|---|----|------|----|----|
| 4 | 60 | 1,75 | 15 | 9  |
| 5 | 60 | 1,70 | 14 | 10 |
| 6 | 60 | 1,70 | 14 | 7  |

$$d^2(4,5) = (60-60)^2 + (1.75-1.70)^2 + (15-14)^2 + (9-10)^2 = 2.00025$$

$$d^2(4,6) = (60-60)^2 + (1.75-1.70)^2 + (15-14)^2 + (9-7)^2 = 5.00000$$

$$d^2(5,6) = (60-60)^2 + (1.70-1.70)^2 + (14-14)^2 + (10-7)^2 = 9.00000$$

Now let's express the height in centimetres

$$d^2(4,5) = (60-60)^2 + (175-170)^2 + (15-14)^2 + (9-10)^2 = 27$$

$$d^2(4,6) = (60-60)^2 + (175-170)^2 + (15-14)^2 + (9-7)^2 = 30$$

$$d^2(5,6) = (60-60)^2 + (170-170)^2 + (14-14)^2 + (10-7)^2 = 9.00000$$

When height is expressed in metres, student 6 is closer to student 4 than to student 5, when it is expressed in centimetres, it is the opposite.

The distance depends on the units of measurement chosen. To stabilise the distance, the usual procedure is to centre and reduce the variables .Let's calculate the distances between students 4,5,6

$$d^2(4,5) = (60-60)^2 / 55.25 + (1.75-1.70)^2 / 0.005525 + (15-14)^2 / 0.56 + (9-10)^2 / 11 = 2.328$$

$$d^2(4,6) = (60-60)^2 / 55.25 + (1.75-1.70)^2 / 0.005525 + (15-14)^2 / 0.56 + (9-7)^2 / 11 = 2.601$$

$$d^2(5,6) = (60-60)^2 / 55.25 + (1.70-1.70)^2 / 0.005525 + (14-14)^2 / 0.56 + (10-7)^2 / 11 = 0.819$$

The distance no longer depends on the measurement units in which the variables are expressed. So we can say that student 6 is closer to student 5 than student 4 regardless of the units of measurement.

## 3.2 method description

We can calculate all the distances between the statistical units using the previous formula: in the case of 10 statistical units, this gives  $10 * 9 / 2$  distances, i.e. 45. In the case of 100 statistical units, we obtain  $100 * 99 / 2$ , i.e. 4950, which is impossible to analyse directly.

PCA describes the data as well as possible, provides an orthonormal axis system preserving all these distances.

### 3.2.1 Definitions

- ✓ The lines closest to the statistical units are called principal axes. Their direction vectors are called principal vectors

**$\Rightarrow$ . The principal vectors, which generate the principal axes, are the eigenvectors of the correlation matrix associated with the eigenvalues**

## Exemple

In this example, we give the correlations between the initial variables, the eigenvalues, and the principal vectors obtained by PCA.

|        | poids      | taille     | age        | note       |
|--------|------------|------------|------------|------------|
| poids  | 1.0000000  | 0.3665158  | 0.4854043  | -0.5678917 |
| taille | 0.3665158  | 1.0000000  | 0.3955146  | -0.6287373 |
| age    | 0.4854043  | 0.3955146  | 1.0000000  | -0.3223292 |
| note   | -0.5678917 | -0.6287373 | -0.3223292 | 1.0000000  |

\$values

```
[1] 2.3908988 0.7503114 0.5844061 0.2743837
```

\$vectors

|      | [,1]       | [,2]       | [,3]       | [,4]       |
|------|------------|------------|------------|------------|
| [1,] | -0.5079730 | 0.3065293  | 0.6593007  | 0.4618721  |
| [2,] | -0.5038351 | -0.4647035 | -0.5253477 | 0.5041931  |
| [3,] | -0.4453032 | 0.7057876  | -0.4712381 | -0.2854882 |
| [4,] | 0.5383481  | 0.4381258  | -0.2593591 | 0.6715355  |

## Remarque

✓ These vectors are unitary and orthogonal

$$\|U1\|^2 = -0.5080^2 + -0.5038^2 + -0.4453^2 + 0.5383^2 \\ = 1$$

$$U1 \cdot U2 = -0.5079730 * 0.3065293 + -0.5038351 * -0.4647035 + -0.4453032 * 0.705787 + 0.5383481 * 0.438125 \\ = 0$$

- ✓ the list of coordinates of the statistical units on the principal axis generated by the principal vector is called the principal component

## Example

- ✓ We calculate the coordinates of statistical unit 1 on the first two principal axes

poids:  $x_1(1)=45$   $m_1=58.5$  ,  $\sigma=7.43303$

$$X'_1(1) = (45 - 58.5) / 7.43303 = -1.816$$

taille:  $x_2(1)=1.50$   $m_2=1.635$  ,  $\sigma=0.07433$

$$X'_2(1) = (1.50 - 1.635) / 0.07433 = -1.816$$

Age:  $x_3(1)=13$   $m_3=13.8$  ,  $\sigma=0.74833$

$$X'_3(1) = (13 - 13.8) / 0.74833 = -1.069$$

Note:  $x_4(1)=14$   $m_4=12$  ,  $\sigma=3.31662$

$$X'_4(1) = (14 - 12) / 3.31662 = 0.603$$

$$c1(1) = -0.5079730 * -1.816 + -0.5038351 * -1.816 + -0.4453032 * -1.069 + 0.5383481 * 0.603 = -2.638$$

$$c2(1) = 0.3065293 * -1.816 + -0.4647035 * -1.816 + 0.7057876 * -1.069 + 0.4381258 * 0.603 = -0.203$$



The following example lists the coordinates of the statistical units on the principal axes obtained by PCA.

|    | Dim.1      | Dim.2       | Dim.3       | Dim.4       |
|----|------------|-------------|-------------|-------------|
| 1  | -2.6383478 | -0.20304035 | -0.10408766 | -1.04443266 |
| 2  | -1.9434521 | -0.35783346 | 0.31559230  | 0.34952125  |
| 3  | -1.4422179 | -0.80252677 | 0.59077968  | 0.48620265  |
| 4  | 2.0830449  | 0.07837172  | 1.20080516  | -0.19195868 |
| 5  | 0.9867493  | -0.42008337 | 0.29589940  | 0.05285961  |
| 6  | 1.4737035  | -0.81638306 | 0.06130029  | -0.55456712 |
| 7  | 1.3169524  | 0.35329157  | -1.45426142 | -0.40902793 |
| 8  | -0.4313990 | -0.13555130 | -1.24948767 | 0.67416087  |
| 9  | -0.5711642 | 2.38554456  | 0.41285443  | 0.07121803  |
| 10 | 1.1661308  | -0.08178953 | -0.06939452 | 0.56602398  |

## Remarque

Each principal component defines a new variable. These principal components have the following properties:

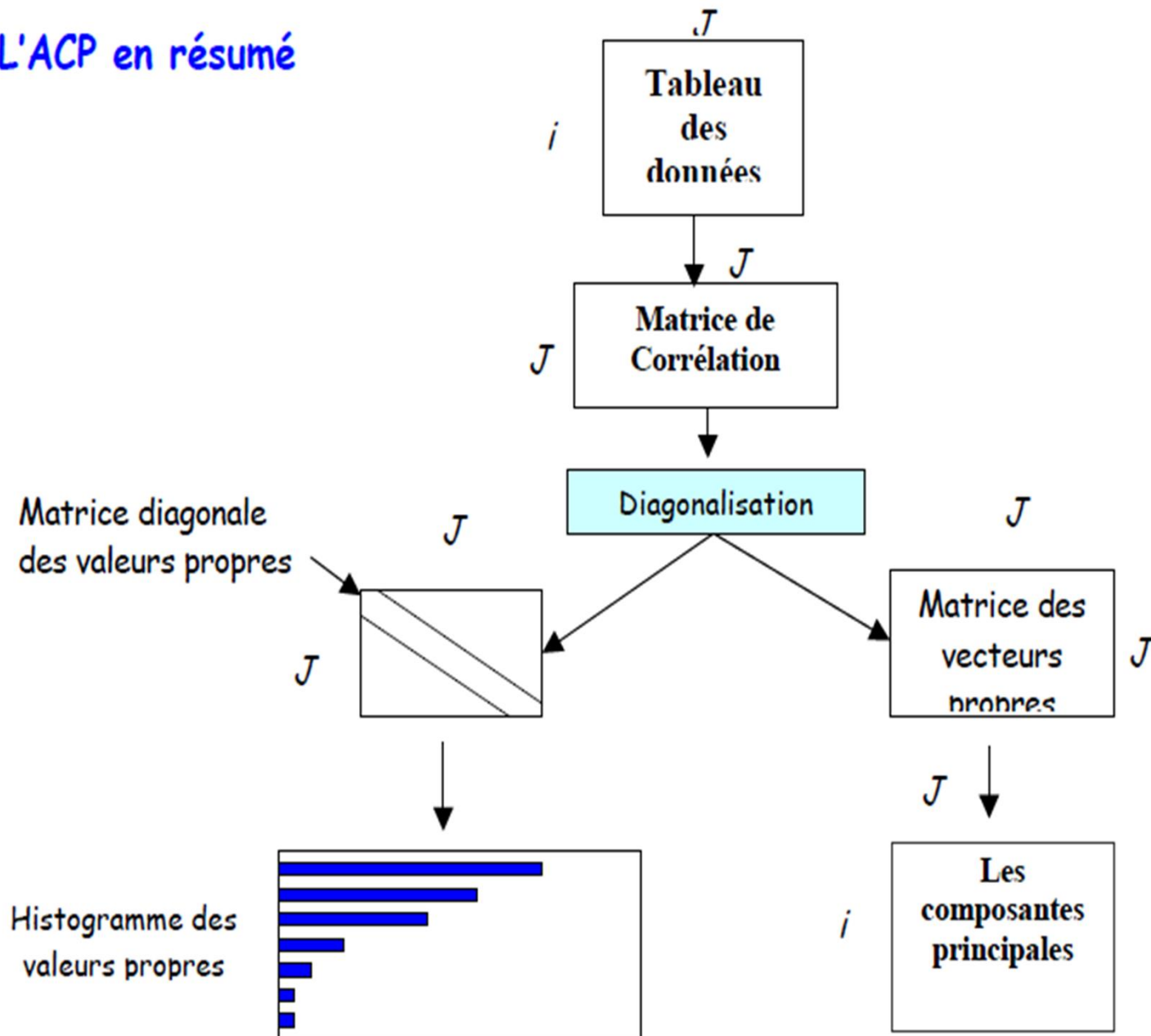
- ✓ the principal components are centred.
- ✓ The principal components are uncorrelated two by two (the correlation coefficients are nuls)

## Remarque

When all axes are considered, all distances are exactly reconstructed

|              | Axe 1 | Axe 1 et 2 | Axe 1 ,2,3 | Axe 1 ,2,3,4 |
|--------------|-------|------------|------------|--------------|
| $d^2(4,5) =$ | 1.201 | 1.449      | 2.268      | 2.328        |
| $d^2(4,6) =$ | 0.371 | 1.170      | 2.470      | 2.601        |
| $d^2(5,6) =$ | 0.237 | 0.394      | 0.449      | 0.819        |

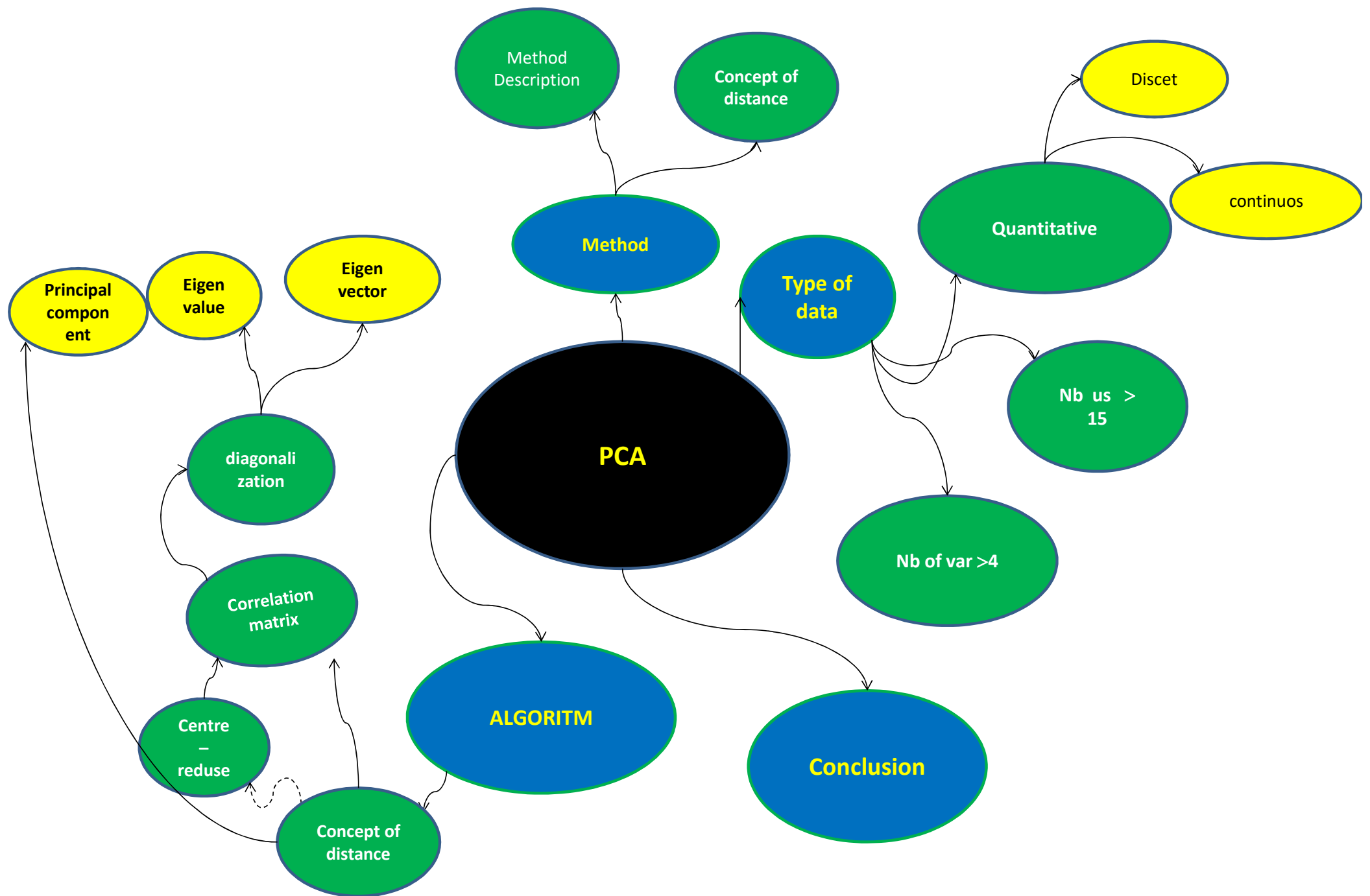
## L'ACP en résumé





## 4. Conclusion

The principal components are very useful in the interpretation: they make it possible to explain the relations between the initial variables and to justify the construction of the homogeneous groups of statistical units



Mind Mapping of PCA